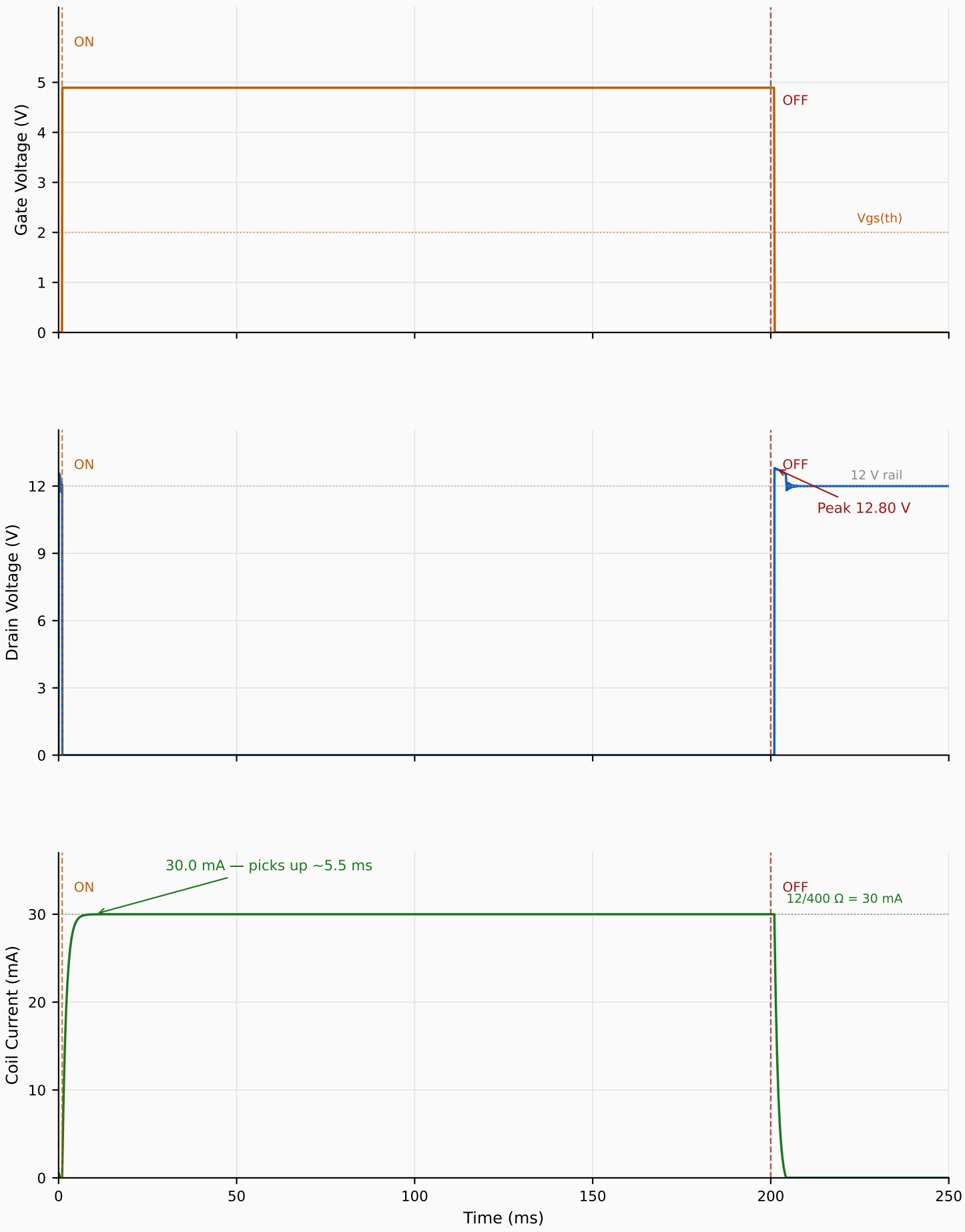
Q2 — Relay Coil Drive (SANYOU SRD-12VDC-SL-C · 400 Ω / 440 mH · 12 V)
Gate Voltage (V)
ON
OFF
Vgs(th)
Drain Voltage (V)
ON
OFF
12 V rail
Peak 12.80 V
Coil Current (mA)
30.0 mA — picks up ~5.5 ms
ON
OFF
12/400 Ω = 30 mA
Time (ms)
BURPG HIP Rev B · LTspice 26.0 · 2026-09-29

Q2 Relay Coil Drive — Simulation Parameters

Parameter	Value	Notes
Flyback spike	**12.80 V**	+0.80 V above 12 V rail — smallest spike of both circuits
Steady coil current	**30.0 mA**	= 12 V / 400 Ω — exact
Gate voltage	**4.89 V**	Same drive as Q1 — fully enhanced
L/R rise time (τ)	**1.1 ms**	440 mH / 400 Ω
Relay pick-up time	**~5.5 ms**	5τ — contacts close well before igniter circuit becomes live
Coil power	**360 mW**	30 mA × 12 V — SANYOU SRD rated for continuous duty
Coil energy @ 45 s	**16.2 J**	360 mW × 45 s — no thermal concern
MOSFET (Q2) power	**<1 mW**	0.03^2 × 22 mΩ — negligible; consider 2N7002 SOT-23 for BOM

VERDICT: PASS. Both relay coil and MOSFET are within thermal limits at any test duration.